Skills Summary

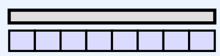
Gaps and Overlaps When We Measure

Use this reference while completing your worksheet or when studying.

Skill 1 — Lay the cubes with no gaps and no overlaps

Rule:

Start at one end. Put the cubes **side by side**: touching, but never on top of each other, and never with a space between. Then the **number of cubes is the length**.



good

Example: A pencil is covered by cubes laid end to end, with no gaps and no overlaps. How long is the pencil?

Step 1. The cubes touch and none sit on top of another, so every cube covers new pencil — that means I may simply count the cubes.

Step 2. Touch each cube once and count on without skipping or repeating; the last cube counted is 6. ⇒ the pencil is **6** cubes long

Try it: A spoon is covered by cubes laid end to end, no gaps and no overlaps, and the last cube counted is 4. How long is the spoon?

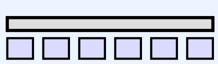
check: 4

Skill 2 — Spot a gap or an overlap mistake

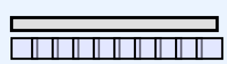
Rule:

Gaps leave bare spaces, so too few cubes reach the end: the count is **too small**.

Overlaps let cubes pile onto each other, so extra cubes squeeze on: the count is **too big**.



gaps: too small



overlaps: too big

Example: Ben left gaps and counted 5. A careful measure gives 7. Write $<$ or $>$: Ben's count ______ real length.

Step 1. Ben left gaps, so I expect his number to be too small — but I still check by comparing the two counts.

Step 2. Ben's count is 5 and the real length is 7; 5 comes first when we count, so it is the smaller number, and the symbol opens toward the bigger one. ⇒ Ben's count $<$ real length

Try it: Lia let her cubes overlap and counted 9. A careful measure gives 6. Write $<$ or $>$: Lia's count ______ real length.

check: $>$

Skill 3 — Fix the count after a gap or overlap

Rule:

The object never changes. Only the lay-down does. To fix a count, take the **careful** length and the **sloppy** count and find **how far apart** they are — count on from the smaller number to the bigger one, and that is how many cubes the gaps hid or the overlaps added.

Example: Ben's gaps: he counted 5, but the careful length is 7. How many cubes did his gaps hide?

Step 1. "How many hidden" asks how far apart the two numbers are, so this is a difference, not a total.

Step 2. Count on from 5 up to 7 — that is two counts, so the difference is 2. ⇒ the gaps hid **2** cubes

Try it: Lia's overlaps: she counted 9, but the careful length is 6. How many extra cubes did the overlapping add?

check: **3**

(!) Watch out: A *bigger number* is not a *better measurement*. Always look at how the cubes were laid before you trust a count — and never add the two counts when a question asks "how many more" or "how many hidden".